



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	CompViz – Interactive Compiler Phase Visualization and Analysis Platform	
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Abstract

Compiler Design involves complex phases such as lexical analysis, syntax analysis, intermediate code generation, optimization, and code generation. Students often find these concepts difficult because compiler operations are sequential and mostly taught theoretically, making transformations such as tokenization, parse trees, intermediate code, and optimization difficult to visualize and understand.

Existing compiler visualizers, tutorials, and learning tools demonstrate individual compiler phases, but many provide static outputs, limited interaction, and insufficient explanations of how one phase transforms into the next. This creates a gap between theoretical concepts and the actual working of a compiler.

To address this, CompVizAI provides an interactive, step-by-step visualization of the complete compilation process. It uses Regex-based lexical analysis, finite automata, CFG, LL (1)/recursive-descent parsing, AST, Three Address Code (TAC), basic blocks, flow graphs, DAG-based optimization, and simple code generation. An AI-based explanation model helps users understand the purpose and transformation performed at each phase.

In the future, the system can be extended with SLR, CLR and LALR parsing, multiple programming languages, advanced optimizations, intelligent error explanations, interactive grammar construction, and personalized AI-based learning recommendations, making it a comprehensive platform for learning compiler concepts.

Signature of the Guide

HOD